11.1 ADC introduction

The 12-bit ADC is a successive approximation analog-to-digital converter. It has up to 18 multiplexed channels allowing it measure signals from sixteen external and two internal sources. A/D conversion of the various channels can be performed in single, continuous, scan or discontinuous mode. The result of the ADC is stored in a left-aligned or right-aligned 16-bit data register.

The analog watchdog feature allows the application to detect if the input voltage goes outside the user-defined high or low thresholds.

The ADC input clock is generated from the PCLK2 clock divided by a prescaler and it must not exceed 14 MHz, refer to *Figure 8* for low-, medium-, high- and XL-density devices, and to *Figure 11* for connectivity line devices.

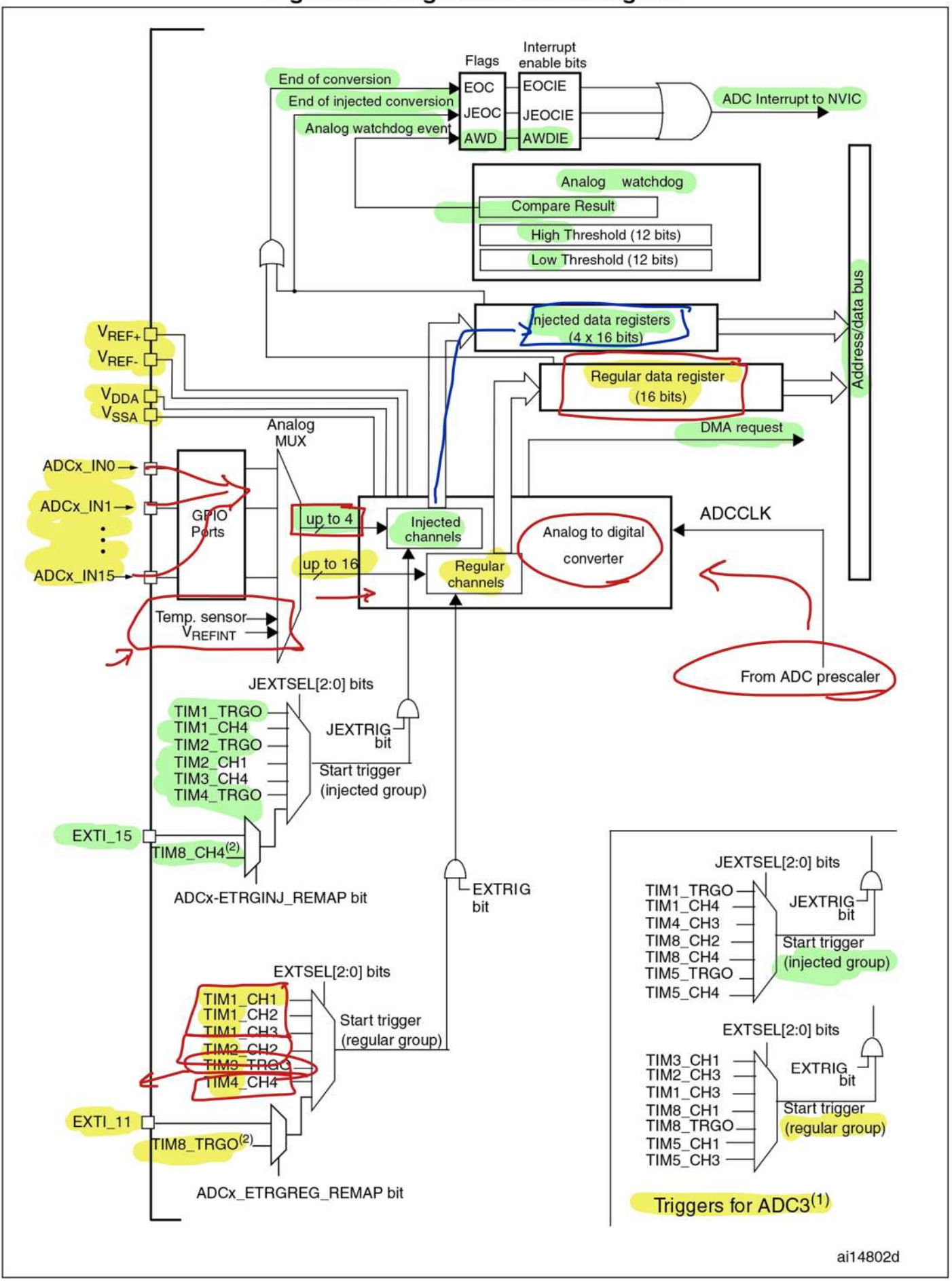
11.2 ADC main features

- 12-bit resolution
- Interrupt generation at End of Conversion, End of Injected conversion and Analog watchdog event
- Single and continuous conversion modes
- Scan mode for automatic conversion of channel 0 to channel 'n'
- Self-calibration
- Data alignment with in-built data coherency
- Channel by channel programmable sampling time
- External trigger option for both regular and injected conversion
- Discontinuous mode
- Dual mode (on devices with 2 ADCs or more)
- ADC conversion time:
  - STM32F103xx performance line devices: 1 µs at 56 MHz (1.17 µs at 72 MHz)
  - STM32F101xx access line devices: 1 µs at 28 MHz (1.55 µs at 36 MHz)
  - STM32F102xx USB access line devices: 1.2 µs at 48 MHz
  - STM32F105xx and STM32F107xx devices: 1 µs at 56 MHz (1.17 µs at 72 MHz)
- ADC supply requirement: 2.4 V to 3.6 V
- ADC input range: $V_{REF-} \leq V_{IN} \leq V_{REF+}$
- DMA request generation during regular channel conversion

The block diagram of the ADC is shown in *Figure 22*.

Note: *V_{REF-}, if available (depending on package), must be tied to V_{SSA}.*

Figure 22. Single ADC block diagram



1. ADC3 has regular and injected conversion triggers different from those of ADC1 and ADC2.
2. TIM8_CH4 and TIM8_TRGO with their corresponding remap bits exist only in High-density and XL-density products.

11.5 Data alignment

ALIGN bit in the ADC_CR2 register selects the alignment of data stored after conversion. Data can be left or right aligned as shown in *Figure 27.* and *Figure 28.*

The injected group channels converted data value is decreased by the user-defined offset written in the ADC_JOFRx registers so the result can be a negative value. The SEXT bit is the extended sign value.

For regular group channels no offset is subtracted so only twelve bits are significant.

Figure 27. Right alignment of data

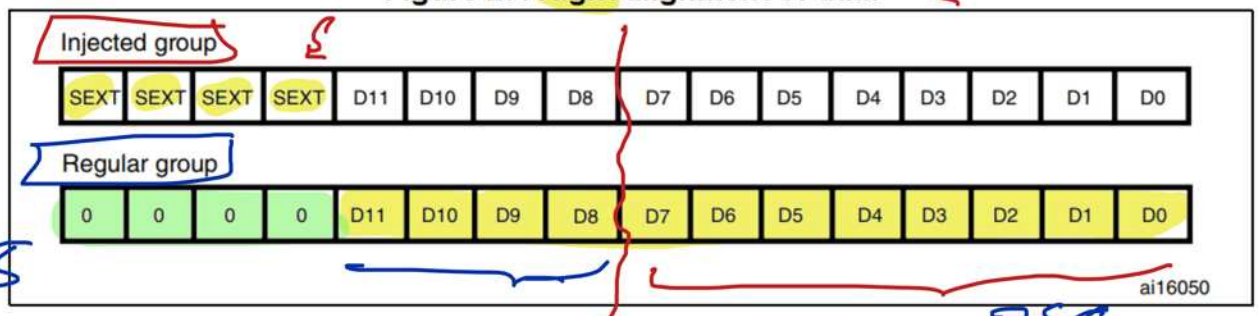
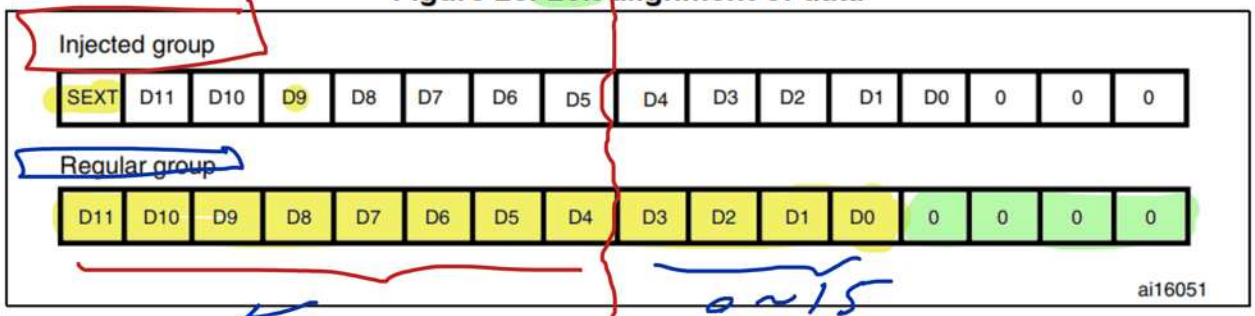


Figure 28. Left alignment of data



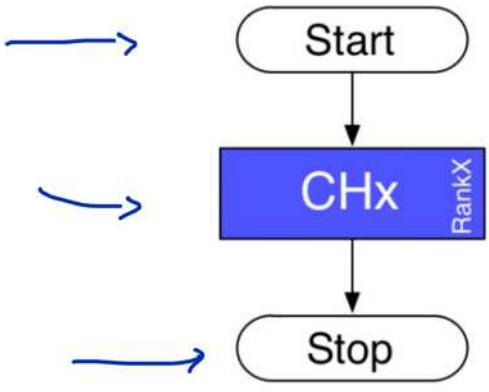


Figure 12.5: Single-channel, single conversion mode

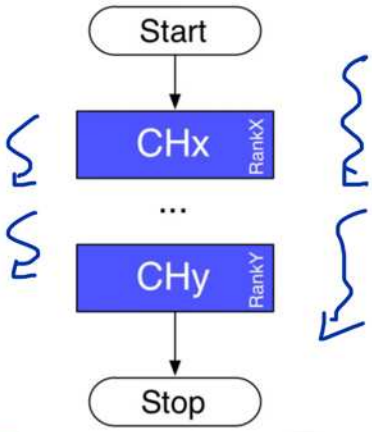


Figure 12.6: Scan single conversion mode

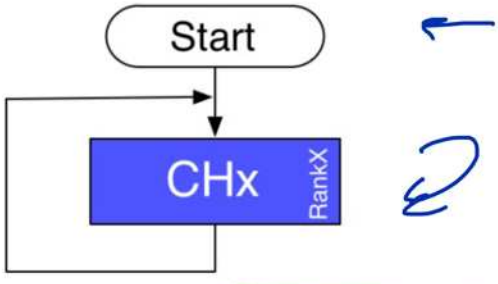


Figure 12.7: Single-channel, continuous conversion

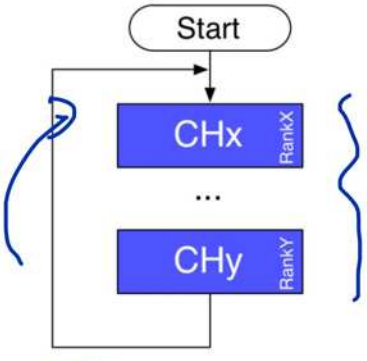
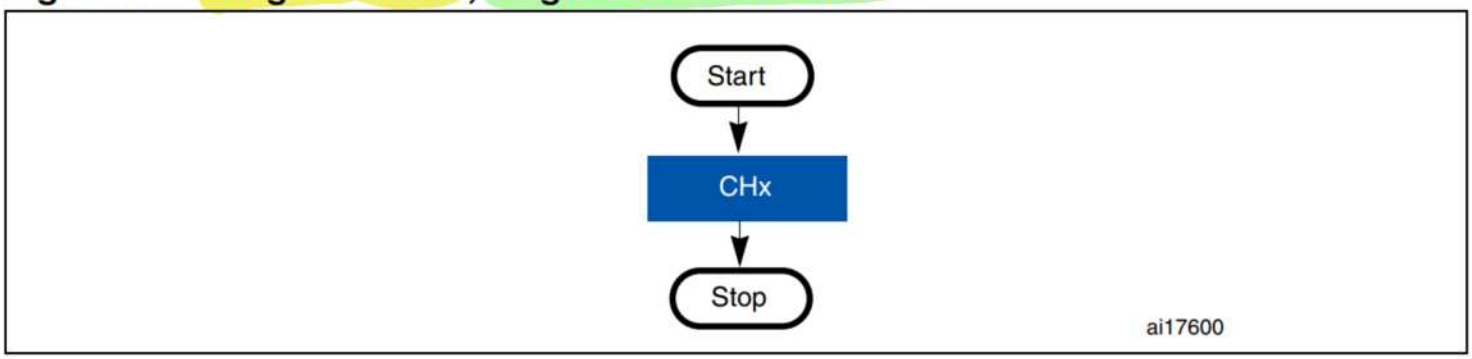


Figure 12.8: Scan continuous conversion mode

Figure 1. Single-channel, single conversion mode



11.6 Channel-by-channel programmable sample time

ADC samples the input voltage for a number of ADC_CLK cycles which can be modified using the SMP[2:0] bits in the ADC_SMPR1 and ADC_SMPR2 registers. Each channel can be sampled with a different sample time.

The total conversion time is calculated as follows:

Tconv = Sampling time + 12.5 cycles

Example:

With an ADCCLK = 14 MHz and a sampling time of 1.5 cycles:

Tconv = 1.5 + 12.5 = 14 cycles = 1 µs

Function name

HAL_StatusTypeDef HAL_ADC_Start (ADC_HandleTypeDef * hadc)

Function description

Enables ADC, starts conversion of regular group.

Parameters

- **hadc:** ADC handle

Return values

- **HAL:** status

`HAL_ADC_PollForConversion`

Function name

HAL_StatusTypeDef HAL_ADC_PollForConversion (ADC_HandleTypeDef * hadc, uint32_t Timeout)

Function description

Wait for regular group conversion to be completed.

Parameters

- **hadc:** ADC handle
- **Timeout:** Timeout value in millisecond.

Return values

- **HAL:** status

Notes

- This function cannot be used in a particular setup: ADC configured in DMA mode. In this case, DMA resets the flag EOC and polling cannot be performed on each conversion.
- On STM32F1 devices, limitation in case of sequencer enabled (several ranks selected): polling cannot be done on each conversion inside the sequence. In this case, polling is replaced by wait for maximum conversion time.

Function name

uint32_t HAL_ADC_GetValue (ADC_HandleTypeDef * hadc)

Function description

Get ADC regular group conversion result.

Parameters

- **hadc:** ADC handle

Return values

- **ADC:** group regular conversion data

Notes

- Reading register DR automatically clears ADC flag EOC (ADC group regular end of unitary conversion).
- This function does not clear ADC flag EOS (ADC group regular end of sequence conversion). Occurrence of flag EOS rising: If sequencer is composed of 1 rank, flag EOS is equivalent to flag EOC.If sequencer is composed of several ranks, during the scan sequence flag EOC only is raised, at the end of the scan sequence both flags EOC and EOS are raised. To clear this flag, either use function: in programming model IT: HAL_ADC_IRQHandler(), in programming model polling: HAL_ADC_PollForConversion() or __HAL_ADC_CLEAR_FLAG(&hadc, ADC_FLAG_EOS).

Function name

HAL_StatusTypeDef HAL_ADC_Stop (ADC_HandleTypeDef * hadc)

Function description

Stop ADC conversion of regular group (and injected channels in case of auto_injection mode), disable ADC peripheral.

Parameters

- **hadc:** ADC handle

Return values

- **HAL:** status.

Notes

- : ADC peripheral disable is forcing stop of potential conversion on injected group. If injected group is under use, it should be preliminarily stopped using HAL_ADCEx_InjectedStop function.

Figure 4. Single-channel, continuous conversion mode

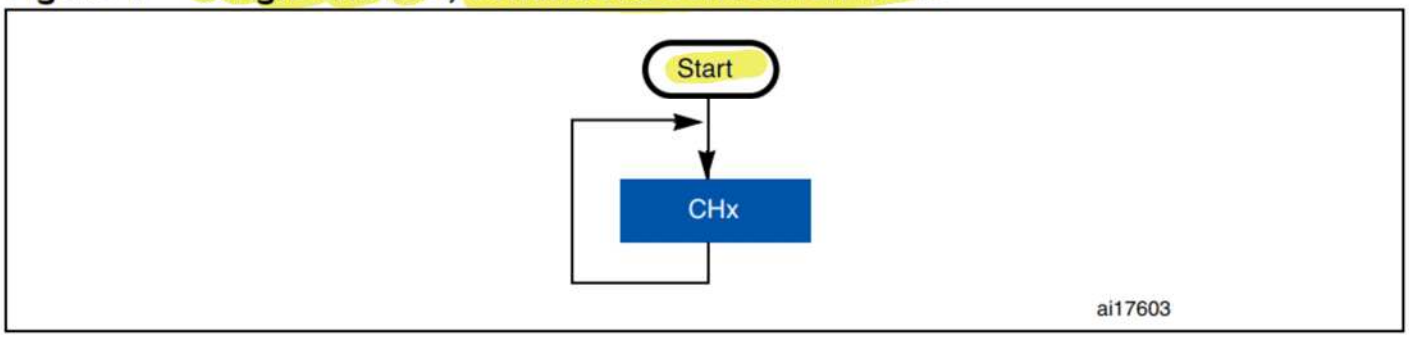


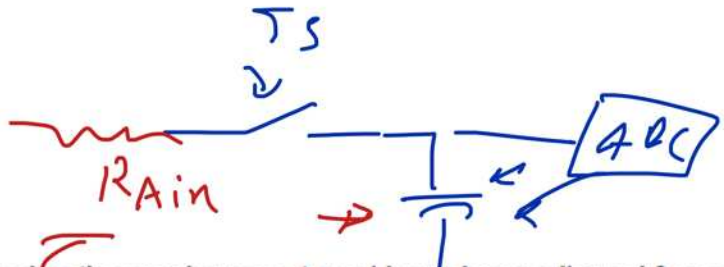
Table 49. ADC accuracy(1) (2) (3)

Symbol	Parameter	Test conditions	Typ	Max(4)	Unit
ET	Total unadjusted error	f_{PCLK2} = 56 MHz, f_{ADC} = 14 MHz, R_{AIN} < 10 kΩ, V_{DDA} = 2.4 V to 3.6 V Measurements made after ADC calibration	±2	±5	LSB
EO	Offset error		±1.5	±2.5	
EG	Gain error		±1.5	±3	
ED	Differential linearity error		±1	±2	
EL	Integral linearity error		±1.5	±3	

1. ADC DC accuracy values are measured after internal calibration.
2. Better performance could be achieved in restricted V_{DD}, frequency and temperature ranges.
3. ADC Accuracy vs. Negative Injection Current: Injecting negative current on any of the standard (non-robust) analog input pins should be avoided as this significantly reduces the accuracy of the conversion being performed on another analog input. It is recommended to add a Schottky diode (pin to ground) to standard analog pins which may potentially inject negative current.
Any positive injection current within the limits specified for $I_{INJ(PIN)}$ and $\Sigma I_{INJ(PIN)}$ in *Section 5.3.12* does not affect the ADC accuracy.
4. Guaranteed based on test during characterization.

Equation 1: R_{AIN} max formula:

$$R_{AIN} < \frac{T_S}{f_{ADC} \times C_{ADC} \times \ln(2^{N+2})} - R_{ADC}$$



The formula above (*Equation 1*) is used to determine the maximum external impedance allowed for an error below 1/4 of LSB. Here N = 12 (from 12-bit resolution).

Table 47. R_{AIN} max for f_{ADC} = 14 MHz(1)

T_S (cycles)	t_S (µs)	R_{AIN} max (kΩ)
1.5	0.11	0.4
7.5	0.54	5.9
13.5	0.96	11.4
28.5	2.04	25.2
41.5	2.96	37.2
55.5	3.96	50
71.5	5.11	NA
239.5	17.1	NA

1. Guaranteed based on test during characterization.

Table 46. ADC characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{DDA}	Power supply	-	2.4	-	3.6	V
V_{REF+}	Positive reference voltage	-	2.4	-	V_{DDA}	V
I_{VREF}	Current on the V_{REF} input pin	-	-	160(1)	220(1)	µA
f_{ADC}	ADC clock frequency	-	0.6	-	14	MHz
f_S(2)	Sampling rate	-	0.05	-	1	MHz
f_{TRIG}(2)	External trigger frequency	f_{ADC} = 14 MHz	-	-	823	kHz
			-	-	17	$1/f_{ADC}$
V_{AIN}(3)	Conversion voltage range		0 (V_{SSA} or V_{REF-} tied to ground)	-	V_{REF+}	V
R_{AIN}(2)	External input impedance	See *Equation 1* and *Table 47* for details	-	-	50	kΩ
R_{ADC}(2)	Sampling switch resistance	-	-	-	1	kΩ
C_{ADC}(2)	Internal sample and hold capacitor	-	-	-	8	pF
t_{CAL}(2)	Calibration time	f_{ADC} = 14 MHz	5.9			µs
		-	83			$1/f_{ADC}$
t_{lat}(2)	Injection trigger conversion latency	f_{ADC} = 14 MHz	-	-	0.214	µs
		-	-	-	3(4)	$1/f_{ADC}$
t_{latr}(2)	Regular trigger conversion latency	f_{ADC} = 14 MHz	-	-	0.143	µs
		-	-	-	2(4)	$1/f_{ADC}$
t_S(2)	Sampling time	f_{ADC} = 14 MHz	0.107	-	17.1	µs
		-	1.5	-	239.5	$1/f_{ADC}$
t_{STAB}(2)	Power-up time	-	0	0	1	µs
t_{CONV}(2)	Total conversion time (including sampling time)	f_{ADC} = 14 MHz	1	-	18	µs
		-	14 to 252 (t_S for sampling +12.5 for successive approximation)			$1/f_{ADC}$

1. Guaranteed based on test during characterization.
2. Guaranteed by design.
3. In devices delivered in VFQFPN and LQFP packages, V_{REF+} is internally connected to V_{DDA} and V_{REF-} is internally connected to V_{SSA}. Devices that come in the TFBGA64 package have a V_{REF+} pin but no V_{REF-} pin (V_{REF-} is internally connected to V_{SSA}), see *Table 5* and *Figure 7*.
4. For external triggers, a delay of $1/f_{PCLK2}$ must be added to the latency specified in *Table 46*.

11.4 Calibration

The ADC has an built-in self calibration mode. Calibration significantly reduces accuracy errors due to internal capacitor bank variations. During calibration, an error-correction code (digital word) is calculated for each capacitor, and during all subsequent conversions, the error contribution of each capacitor is removed using this code.

Calibration is started by setting the CAL bit in the ADC_CR2 register. Once calibration is over, the CAL bit is reset by hardware and normal conversion can be performed. It is recommended to calibrate the ADC once at power-on. The calibration codes are stored in the ADC_DR as soon as the calibration phase ends.

Note: *It is recommended to perform a calibration after each power-up.*

Before starting a calibration, the ADC must have been in power-on state (ADON bit = '1') for at least two ADC clock cycles.

8.2.3 Detailed description of functions

`HAL_ADCEx_Calibration_Start`

Function name

HAL_StatusTypeDef HAL_ADCEx_Calibration_Start (ADC_HandleTypeDef * hadc)

Function description

Perform an ADC automatic self-calibration Calibration prerequisite: ADC must be disabled (execute this function before HAL_ADC_Start() or after HAL_ADC_Stop()).

Parameters

- **hadc:** ADC handle

Return values

- **HAL:** status

`HAL_ADC_Start_DMA`

Function name

HAL_StatusTypeDef HAL_ADC_Start_DMA (ADC_HandleTypeDef * hadc, uint32_t * pData, uint32_t Length)

Function description

Enables ADC, starts conversion of regular group and transfers result through DMA.

Function name

void HAL_ADC_ConvCpltCallback (ADC_HandleTypeDef * hadc)

Function description

Conversion complete callback in non blocking mode.

Parameters

- **hadc:** ADC handle

`HAL_ADC_ConvHalfCpltCallback`

Function name

void HAL_ADC_ConvHalfCpltCallback (ADC_HandleTypeDef * hadc)

Function description

Conversion DMA half-transfer callback in non blocking mode.

Parameters

- **hadc:** ADC handle

Return values

- **None:**